

Program 5: Demonstrate CI Integration by Running Jenkins in a Docker Container

Project Structure

DEVOPSLAB

|--prg5

|--Dockerfile

|--docker-compose.yml

Step 1: Ensure Docker service is running

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ mkdir prg5
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB$ cd prg5
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ docker --version
Docker version 29.1.2, build 890dcca
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ docker compose version
Docker Compose version v5.0.0
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ nano Dockerfile
```

Step 2: Create a Dockerfile and add similar contents



The screenshot shows a terminal window with the title bar "1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/prg5". The editor is GNU nano 7.2, editing a file named "Dockerfile". The content of the Dockerfile is as follows:

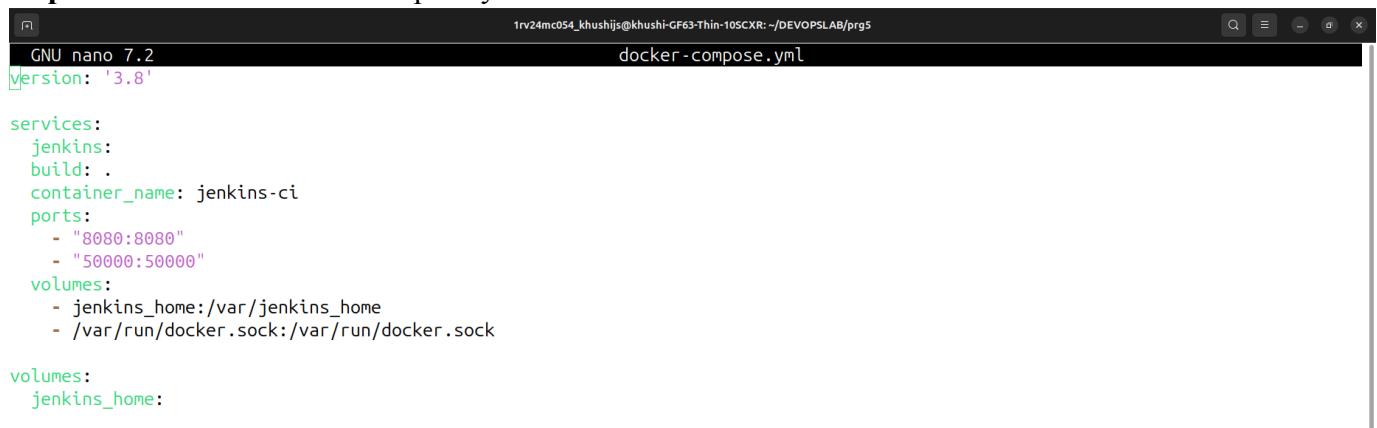
```
FROM jenkins/jenkins:lts

#Switch to root user to install tools
USER root

#Install Docker inside Jenkins container
RUN apt-get update && apt-get install -y docker.io

#Switch back to Jenkins user
USER jenkins
```

Step 3: Make a file docker-compose.yml



The screenshot shows a terminal window with the title bar "1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/prg5". The editor is GNU nano 7.2, editing a file named "docker-compose.yml". The content of the docker-compose.yml file is as follows:

```
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 4: Build and run Jenkins

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR: ~/DEVOPSLAB/prg5
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ nano Dockerfile
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ nano docker-compose.yml
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ docker build -t program-5 .

[+] Building 528.9s (7/7) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.5s
=> => transferring dockerfile: 250B                                0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lt      7.2s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io    0.0s
=> [internal] load .dockerignore                                   0.3s
=> => transferring context: 2B                                       0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lt@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 74.3s
=> => resolve docker.io/jenkins/jenkins:lt@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 0.2s
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 392B / 392B 2.0s
=> => sha256:986be6969b0a04975a2e559b30a96d766bf360d0b340b2789b611688ba8bbea 1.29kB / 1.29kB 2.2s
=> => sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de53927 2.23kB / 2.23kB 2.0s
=> => sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bf6d6899 72.05MB / 72.05MB 62.7s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acb06245 6.29MB / 6.29MB 7.6s
=> => sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 189B / 189B 1.3s
=> => sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627 95.32MB / 95.32MB 58.9s
=> => sha256:e0e73f34dee9332ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B 1.8s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB 1.5s
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB 5.9s
=> => sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 65.08MB / 65.08MB 51.3s
=> => extracting sha256:5687d4df0728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 2.6s
=> => extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 0.5s
=> => extracting sha256:68cdef48009293f6a12f5c50b91fcfe4194bd6293334d8a31a45b1ed63cb91a5 0.2s
=> => extracting sha256:e0e73f34dee9332ae4db1d44250aab1f61011a2b7ec9549ff1f6df342c31168 0.3s
=> => extracting sha256:687044b042c5d6d57e2ce207c000ee159f2382cb84cb8d80688c35e7f1ed8627 1.0s
=> => extracting sha256:501deb283b89e30fd3f1f4e27a635bcc1739485e8ba805034552e3d4eaac0b98 0.2s
=> => extracting sha256:a9230f7c9c31780c6b1dc4f2aa7e1073df2fdf88e9c009f6874e7221acb06245 1.2s
=> => extracting sha256:d7dd2e63023897146ec4b913a460382968fbd1cc3682e36f650727a1bf6d6899 1.4s
=> => extracting sha256:3fb70446889ebcc58e91f43f90aafa314798941c318ce016ea62068a3de53927 0.2s

=> => unpacking to docker.io/library/program-5:latest            4.3s
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg5$ docker run -p 8080:8080 program-5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-12-22 14:24:46.238+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2025-12-22 14:24:46.840+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-12-22 14:24:46.878+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea; jvm 21.0.9+10-LTS
2025-12-22 14:24:47.593+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2025-12-22 14:24:47.638+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2025-12-22 14:24:48.000+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2025-12-22 14:24:48.112+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@3c782d8e{Jenkins v2.528.3,,b=file:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9{STARTED}}
2025-12-22 14:24:48.122+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac{HTTP/1.1,(http/1.1)}{0.0.0.0:8080}
2025-12-22 14:24:48.130+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4{STARTING}[12.0.25,sto=0] @2684ms
2025-12-22 14:24:48.130+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2025-12-22 14:24:48.568+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.3
2025-12-22 14:24:49.429+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-12-22 14:24:49.490+0000 [id=53] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2025-12-22 14:24:50.176+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2025-12-22 14:24:50.180+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-12-22 14:24:50.181+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-12-22 14:24:50.846+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-12-22 14:24:50.847+0000 [id=58] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-12-22 14:24:50.848+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-12-22 14:24:50.852+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-12-22 14:24:50.921+0000 [id=75] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
```

Step 5: Test the running of Jenkins by logging into

